

VentureGraph — One-Page Handout

VentureGraph 2.0 · The FinTech Foundry Edition

C-DE422 · Big Data Engineering II · Egypt University of Informatics
Mohamed Hares · May 2026

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Identifying Temporal Precursor Investors & Smart Money Silence in Co-Investment Networks

The Question

Classical link prediction scores edges that **form**. This project scores edges that were **expected to form but did not** — and tests whether that absence predicts public-market sector returns.

The Data

Crunchbase 2013 snapshot · **12,419 investments** · **3,548 companies** · **4,035 investors** · **5 sectors** · 2000–2013. Directed, time-weighted co-investment graph. Pruned: **1,101 nodes** · **3,870 edges** · density 0.0032 · clustering 0.5147.

The Three Novel Constructs

Construct	Formula	What it measures	File
TPS — Temporal Precursor Score	$\frac{1}{\sum_{c \in \mathcal{T}} \sum_{B \in \mathcal{B}(c)} w(A \rightarrow B, c) \cdot \text{rank}^{-1}(B)}$	Prescience (volume-independent)	<code>tps.py</code>
SSI — Swarm Signal Intensity	TPS-weighted rolling new-entrant inflow	Prescient capital inflow	<code>compute_ssi.py</code>
SMS — Smart Money Silence ■	rate at which top-30% TPS Series A investors decline to follow-on to Series B	Structural absence as signal	<code>sms_engine.py</code>

■ = **primary innovation** for the 5-mark bonus. First operationalization of **inverse link prediction** as a financial signal, extending Lü–Zhou (2011).

The Findings

- **TPS ≠ classical centrality**: rank correlation with out-degree $\rho \approx 0.31$ (low).
- **Louvain communities**: 35 partitions, $Q = 0.473$ — clean separation of "seed scouts" from "elite follow-on."
- **H3 empirical test** (pre-registered): SMS → FF5 sector alpha.

Horizon	Pearson r	p-value	n
k=6	+0.023	0.90	35
k=12	−0.036	0.77	73
k=18	+0.045	0.79	37
k=24	−0.060	0.61	76
k=36	+0.047	0.70	67

Honest finding: H3 is not rejected by the null. Power analysis → needs $n \approx 2,170$ (30× larger). The *methodological* contribution is the deliverable; empirical resolution is explicit future work.

The Audit Infrastructure (rare even in peer-reviewed finance)

- **Pre-registered protocol** (`preregistration.py`): SHA-256-locked hyperparameters + hypotheses + robustness tests before any back-test.

- **Expanding-window firewall** (`expanding_window_tps.py`): per-snapshot SHA-256 audit, prevents look-ahead contamination.
- **Two-way clustered SE** (Cameron-Gelbach-Miller) · **wild-cluster bootstrap** (Rademacher, n=999) · **placebo permutation**.

The Dashboard (Part F)

`streamlit run app.py` · Two modes: **Academic** (submission) · **Sovereign** (commercial-direction prototype). 10 interactive quant modules: percolation stress, k-core, spectral gap, dynamic PageRank, Adamic-Adar, Shannon entropy, clique detection, assortativity, triadic closure, information arbitrage.

The Deliverables Map (C-DE422 Brief Compliance)

Brief	File	Status
Part A — Network statistics	<code>graph_construction.py</code> , <code>sna_metrics.py</code>	✓
Part B — Centrality	<code>tps.py</code> , <code>centrality_comparison.py</code>	✓
Part C — Community	<code>community_detection.py</code>	✓
Part D — Advanced analysis	<code>link_prediction.py</code> , <code>sms_engine.py</code> ■	✓
Part E — Visualizations	9 PNGs	✓
Part F — Dashboard	<code>app.py</code> (dual-mode)	✓
Report	<code>REPORT.md</code> + <code>methodology_appendix.md</code>	✓
Bonus (5 marks)	SMS + pre-registration + audit stack	✓

Code: github.com/mohamedhares/venturegraph (*publishing*) · **Paper draft:** `paper_abstract.md` · **Contact:** mohamed.hares@eui.edu.eg